

BACKLOG BUDDY ? STUDY MATERIAL

Power Electronics ? Unit-wise Lecture Notes

Subject Code: EE402 Semester: 4

.....
Condensed notes covering all five units: Power Semiconductor Devices, Controlled Rectifiers, Choppers, Inverters, AC Voltage Controllers.

Unit 1: Power Semiconductor Devices

- Definition, key concepts and significance of Power Semiconductor Devices
- Important diagrams / derivations / algorithms for Power Semiconductor Devices
- Solved examples and previous-year questions on Power Semiconductor Devices

Unit 2: Controlled Rectifiers

- Definition, key concepts and significance of Controlled Rectifiers
- Important diagrams / derivations / algorithms for Controlled Rectifiers
- Solved examples and previous-year questions on Controlled Rectifiers

Unit 3: Choppers

- Definition, key concepts and significance of Choppers
- Important diagrams / derivations / algorithms for Choppers
- Solved examples and previous-year questions on Choppers

Unit 4: Inverters

- Definition, key concepts and significance of Inverters
- Important diagrams / derivations / algorithms for Inverters
- Solved examples and previous-year questions on Inverters

Unit 5: AC Voltage Controllers

- Definition, key concepts and significance of AC Voltage Controllers
- Important diagrams / derivations / algorithms for AC Voltage Controllers
- Solved examples and previous-year questions on AC Voltage Controllers

Exam Tips:

- * Start with the latest syllabus and exam pattern ? focus on units with the highest weightage first.
- * Solve at least 3 previous year papers under timed conditions before the exam.
- * Focus on frequently repeated topics ? they are strong preparation priorities based on available papers (not guarantees).
- * Revise definitions, diagrams and standard derivations ? they earn easy marks.
- * Practise numerical problems from every unit; papers usually include 2-3 compulsory problems.

Disclaimer: Sample study material for preparation support only.

Appearance counts are based on available papers and are not exam predictions.